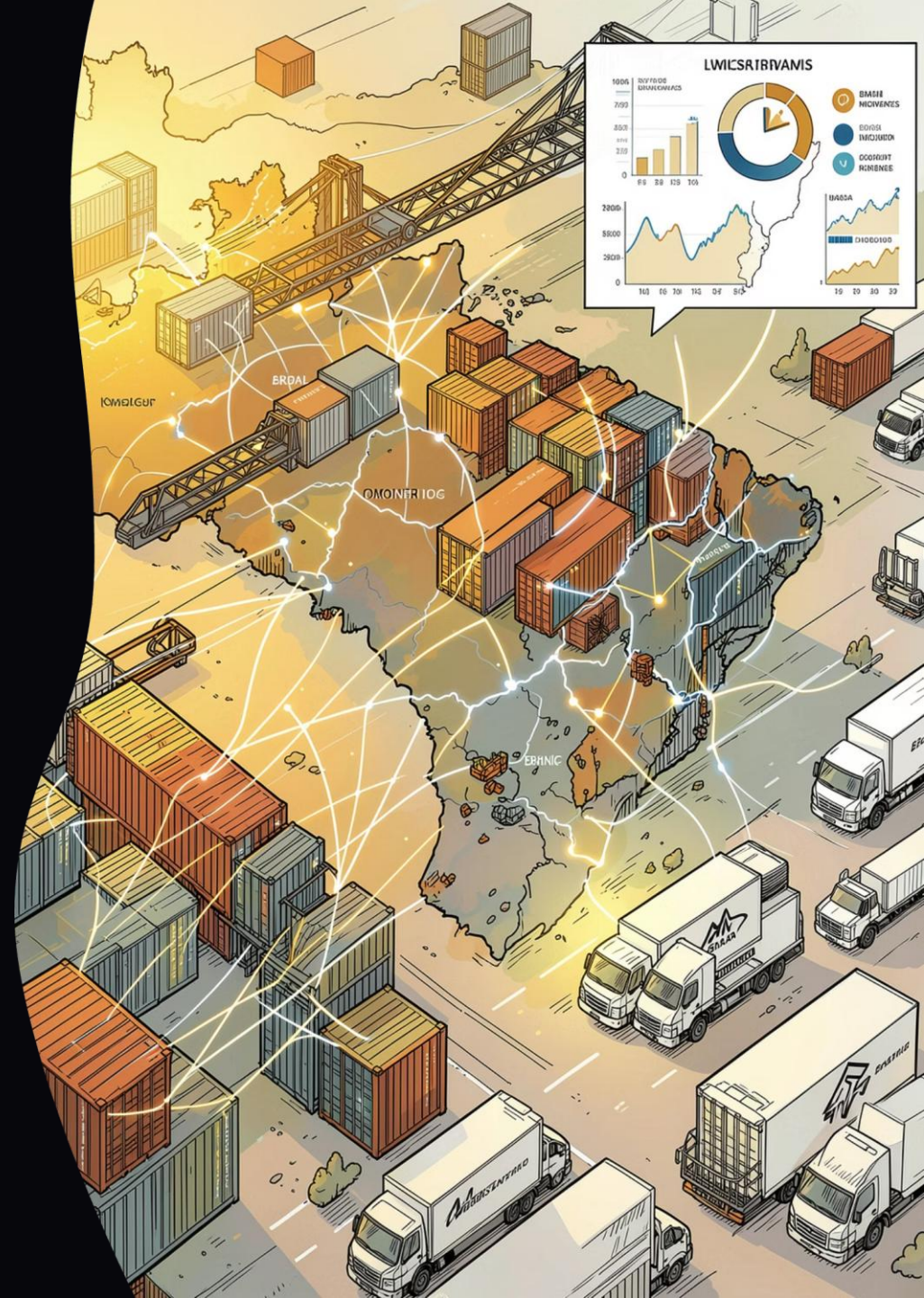


Market Expansion Strategy

Identifying High-Potential Markets for Business Expansion Using SQL, Market Analytics & Power BI for OLIST ECommerce

Prepared by **Swastik Tiwari**



Executive Summary

The company intends to expand into new Brazilian markets but currently lacks a structured, data-driven framework for selecting optimal expansion locations. This analysis bridges that gap with a rigorous, analytics-led methodology.

Objectives

- Identify high-potential states
- Reduce expansion risk
- Maximise ROI
- Improve customer acquisition
- Support executive decision-making

Deliverables

SQL Data Warehouse — Centralised, cleansed repository for all market data.

Customer Segmentation — RFM-based grouping for targeted strategy.

Geographic Analysis — State-level mapping of demand and revenue density.

Financial Scenario Modelling — Bull, Base, and Bear cases with expected value.

Business Understanding

Before expanding into Brazil's diverse regional markets, leadership must confront critical knowledge gaps and answer foundational strategic questions that will shape investment decisions.



Current Challenges

- Unknown market attractiveness across states
- Limited investment budget requiring prioritisation
- Competitive uncertainty in regional landscapes
- Customer distribution imbalance



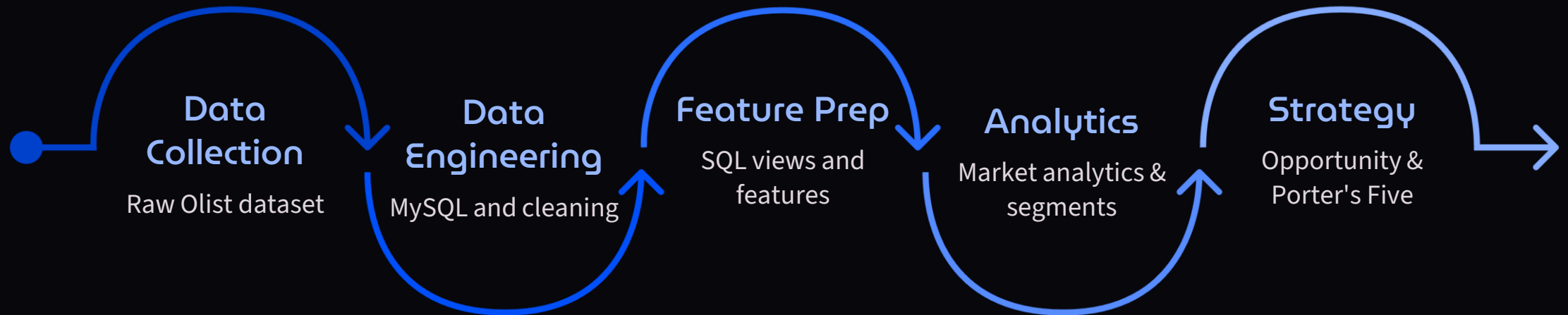
Business Questions

- Which market should be entered first?
- Which state offers the highest ROI?
- How should investment be phased over time?
- Which customer segments should be targeted?



Project Roadmap

A systematic, end-to-end pipeline from raw Olist data through SQL engineering and multi-layered analytics to a final, board-ready business recommendation.



Data Engineering & SQL Pipeline

Leveraging the Olist e-commerce dataset, a robust SQL pipeline was constructed to transform raw transactional data into analysis-ready views. This foundational layer ensured data integrity and enabled advanced downstream analytics.

Dataset Tables

Customers

Orders

Products

Payments

Reviews

Sellers

Key SQL Views Created

Master Transactions

Unified table joining orders, items, payments, and reviews for 360° transaction visibility.

Geolocation Average

Aggregated customer coordinates to approximate regional centres for mapping.

Regional Segment Summary

State-level KPI roll-ups enabling comparative market attractiveness scoring.

SQL work included handling missing values, duplicate detection, data validation, and building joins, views, and indexes to optimize query performance across the warehouse.

Analytics Framework

A structured, six-stage Market Expansion Framework systematically progresses from granular customer behaviour to a holistic strategic recommendation, ensuring every dimension of market viability is quantified.

01

Customer Analytics

RFM segmentation and cohort analysis to identify high-value customer groups.

02

Geographic Analytics

Mapping regional demand and state-wise revenue distribution across Brazil.

03

Market Potential

Estimating TAM, SAM, and SOM to size the addressable opportunity.

04

Competition Analysis

Porter's Five Forces to assess structural attractiveness and rivalry.

05

Financial Analysis

Scenario modelling with Bull, Base, and Bear cases for ROI projection.



CUSTOMER ANALYTICS

Customer Analytics

Using RFM (Recency, Frequency, Monetary) analysis, customers were segmented into strategically actionable groups. Cohort analysis and Customer Lifetime Value (CLV) calculations further illuminated regional customer quality and retention patterns.

Champions

Recent, frequent, high-spend customers driving top-line revenue.

Loyal

Consistent purchasers with strong brand affinity and steady CLV.

At Risk

Previously valuable customers showing declining engagement signals.

Needs Attention

Infrequent buyers with potential to move up or churn depending on intervention.

Key Findings: Customer Lifetime Value varies significantly by region, with retention rates highest among the Loyal and Champion segments. Targeted reactivation campaigns for At Risk customers present the most immediate ROI opportunity.

Geographic & Market Analysis

Geographic analytics revealed pronounced regional imbalances in customer distribution, revenue concentration, and market density. State-wise revenue mapping and TAM/SAM/SOM sizing identified three standout candidates for near-term expansion.

Why These States Were Shortlisted

Qoiás

Emerging central market with low competitive saturation and rising e-commerce adoption.

Minas Qerais

Large population centre, strong revenue density, and underserved logistics network.

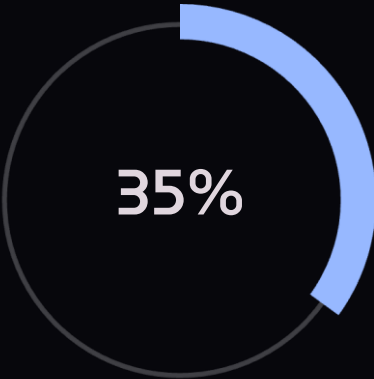
Rio Qrande do Sul

High customer quality scores, robust purchasing power, and favourable SOM-to-TAM ratio.



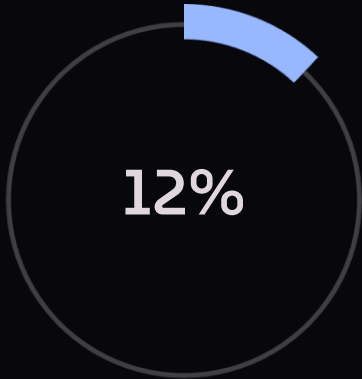
TAM

Total Addressable Market — all e-commerce demand across Brazil.



SAM

Serviceable Available Market — regions reachable by current logistics.

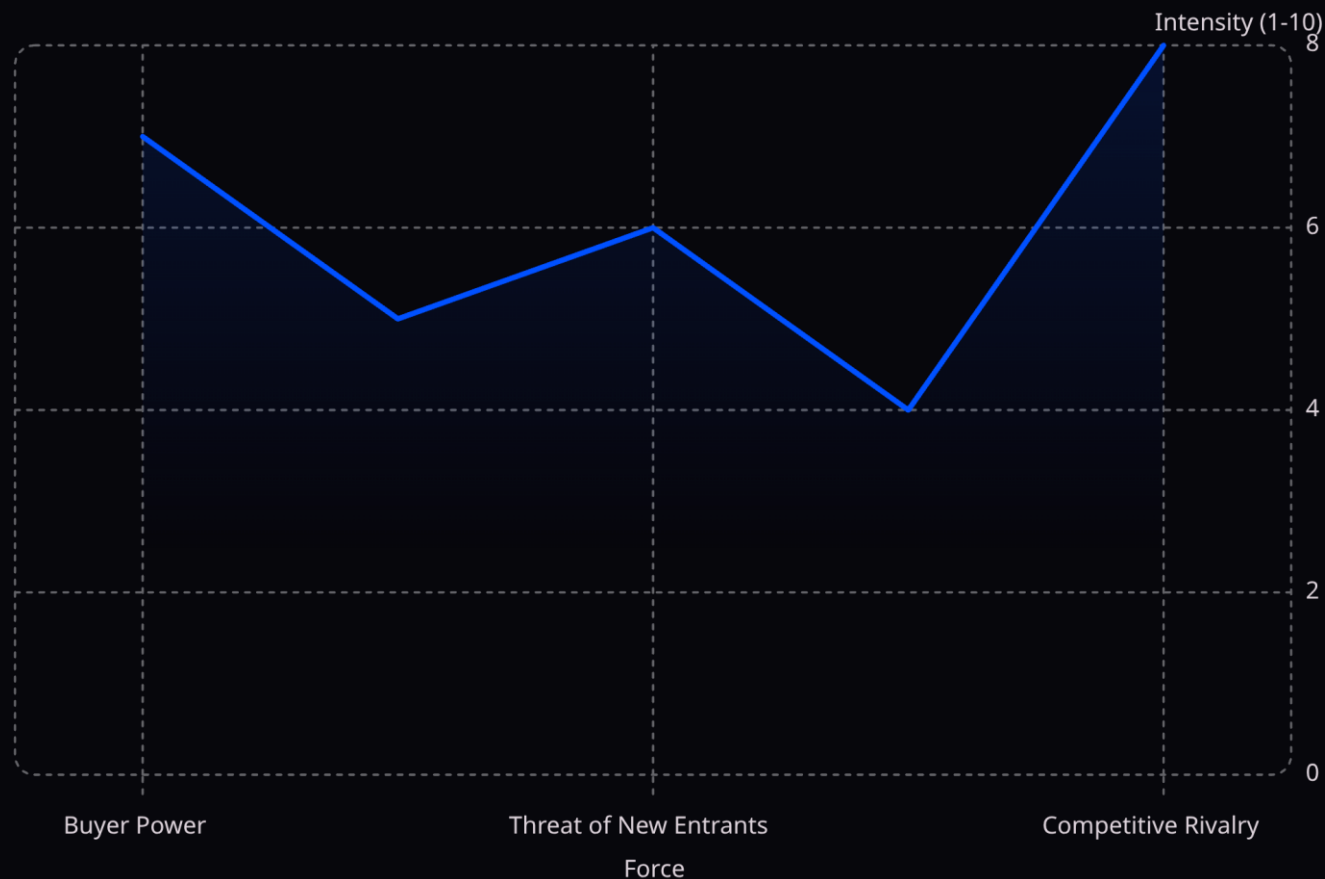


SOM

Serviceable Obtainable Market — realistic near-term capture share.

Market Opportunity Assessment

Porter's Five Forces analysis was applied to each shortlisted state to evaluate structural market attractiveness beyond headline revenue figures. The radar chart below summarises the competitive intensity across all five forces.



Strategic Interpretation

- Rivalry (8/10):** High competition in saturated hubs like São Paulo and Rio de Janeiro favours expansion into emerging states.
- Buyer Power (7/10):** Informed, price-sensitive customers necessitate differentiation through service quality and logistics speed rather than price wars.
- Substitutes (4/10):** Lower threat of substitutes in e-commerce for the target product categories, offering a window for first-mover advantage in underpenetrated states.

Goiás Financial Scenario Analysis

Evaluating Goiás as the primary expansion market, we've assessed profitability, risk, and projected returns across varying market conditions to confirm its strategic potential.

Key Investment Metrics

\$3M

Total Investment

\$60M

Serviceable Obtainable Market (SOM)

23%

Expected ROI

70%

Positive Return Probability

15%

Platform Take Rate

Scenario Analysis

Bull Case: Highest Revenue & Profit

Strong customer adoption and rapid market penetration generate excellent returns, exceeding initial projections.

Base Case: Positive ROI

Moderate customer acquisition aligns with historical growth, making expansion financially attractive and sustainable.

Bear Case: Manageable Losses

Weaker-than-expected demand reduces profitability, but losses remain controllable due to phased investment and prudent cost management.

Financial Insights:

- Goiás presents the highest expected ROI among all shortlisted markets.
- The market demonstrates a strong balance between significant size and manageable competitive intensity.
- A 70% probability of generating positive returns underscores the robustness of the financial model.
- Overall, Goiás represents the best risk-adjusted investment opportunity for immediate expansion.

📌 **Recommendation:** Proceed with immediate expansion into Goiás. Implement a phased investment strategy, closely monitoring customer acquisition and operational performance to optimise returns.

Minas Gerais Financial Scenario Analysis

Assess whether the largest addressable market provides sufficient financial returns despite increased competitive pressure.

Scenario Analysis

Scenario	Outcome	Interpretation
Bull Case	Very high revenue	Large market size enables substantial upside when customer acquisition is successful.
Base Case	Moderate positive ROI	Expansion remains profitable but requires effective competitive positioning.
Bear Case	Negative ROI	High competition may reduce profitability if market penetration is slower than expected.

Key Investment Metrics

\$300

Investment

\$5900

Serviceable Obtainable Market (SOM)

20%

Expected ROI

70%

Positive Return Probability

15%

Platform Take Rate

 **Recommendation:** Position Minas Gerais as the second expansion phase. Invest after establishing operational success in Goiás.

Financial Insights:

- Largest Total Addressable Market (TAM).
- Strong long-term revenue potential.
- Higher competitive risk than Goiás.
- Suitable for scaling after validating the first expansion.

Rio Grande do Sul Financial Scenario Analysis

Evaluate whether Rio Grande do Sul provides sufficient diversification benefits and sustainable financial performance for long-term expansion.

 **Recommendation:** Delay expansion into Rio Grande do Sul until later phases. Consider a smaller pilot program before committing to a full-scale rollout.

Key Investment Metrics

\$3M

Investment

\$54M

Serviceable Obtainable Market (SOM)

10%

Expected ROI

70%

Positive Return Probability

15%

Platform Take Rate

Financial Insights:

- Lowest expected ROI among the three markets.
- Stable customer demand supports gradual expansion.
- Better suited as a diversification strategy rather than an immediate priority.
- Investment should follow successful execution in the higher-ranked markets.

Scenario Analysis

Scenario	Outcome	Interpretation
Bull Case	Positive returns	Strong regional demand can produce attractive long-term profitability.
Base Case	Limited ROI	Expansion is profitable but generates lower returns than other shortlisted markets.
Bear Case	Significant loss	High competitive pressure and slower adoption increase investment risk.

Market Ranking

A comparative analysis of the shortlisted states reveals their distinct strengths and strategic fit for phased market entry, balancing immediate growth with long-term potential.

Rank	State	Key Advantages
1	Goiás	• High Serviceable Obtainable Market (SOM) • Moderate Competitive Saturation • Highest Expected ROI potential in Bull Case • Emerging e-commerce adoption & customer base
2	Minas Gerais	• Largest Total Addressable Market (TAM) • Significant Population Centre • High Revenue Density • Underserved Logistics Network (opportunity)
3	Rio Grande do Sul	• Stable Customer Demand & High Quality Scores • Strong Purchasing Power • Favourable SOM-to-TAM Ratio • Potential for Long-Term Sustainable Growth

Implementation Roadmap: 12-Month Timeline

Our phased 12-month roadmap outlines key milestones from initial market entry to full regional expansion, ensuring sustainable growth and continuous optimisation.



Strategic Directives for Expansion

Our analysis culminates in a set of actionable recommendations, phased by timeline and segmented by business function, designed to maximise return on investment and secure market share.

	Immediate Expand into Goiás, leveraging its emerging market status and low competitive saturation for a quick entry.
	Medium Term Expand into Minas Gerais, capitalising on its large population, high revenue density, and an underserved logistics network.
	Long Term Expand into Rio Grande do Sul, securing long-term sustainable growth with its high customer quality scores and robust purchasing power.
	Marketing Target Loyal Customers with exclusive offers and loyalty programmes to drive repeat purchases and referrals, increasing CLV.
	Retention Implement targeted reactivation campaigns for "Needs Attention" customers to reduce churn and recover lost value from at-risk segments.
	Operations Optimise logistics networks and delivery infrastructure in new states to ensure rapid and reliable service, addressing underserved areas.
	Pricing Differentiate in competitive markets through value-added services rather than price wars, focusing on service quality and speed.

Estimated Business Impact

Implementing the proposed market expansion strategy is projected to yield significant benefits across key operational and financial areas, driving sustainable growth and enhancing market position.



Higher Revenue

Achieve substantial sales growth by tapping into new, underpenetrated markets and expanding the customer base.



Higher Customer Acquisition

Leverage regional opportunities to efficiently acquire new customers, supported by targeted marketing and optimised logistics.



Improved Market Penetration

Increase overall market share and brand presence within the identified high-potential states.



Better ROI

Deliver a favourable return on investment by prioritising strategic market entries with strong growth prospects.



Lower Expansion Risk

Mitigate operational and competitive risks through phased implementation and data-driven market selection.



Improved Decision Making

Enhance strategic agility with a robust analytics framework and executive dashboard for real-time insights.

Project Conclusion

This project successfully developed a comprehensive data-driven market expansion framework, seamlessly integrating SQL, advanced Market Analytics, and Power BI for actionable insights and strategic decision-making.

Key Achievements

- Built a complete, robust SQL data warehouse.
- Engineered analytical views for diverse business needs.
- Developed comprehensive customer segmentation models.
- Performed in-depth market opportunity analysis.
- Conducted detailed financial scenario analysis and stress-testing.
- Identified optimal expansion markets with highest ROI potential.

Final Recommendation

1

Phase 1: Expand into Goiás

2

Phase 2: Expand into Minas Gerais

3

Phase 3: Expand into Rio Grande do Sul